

Introduction



Document Signature App — Java Tech Stack



Introduction

The **Document Signature App** is a secure, enterprise-grade web application that enables users to **upload documents, place digital signatures, share signing links, and generate legally traceable signed PDFs**.

Built using the **Java ecosystem**, this project mirrors how **real-world enterprise document management systems** (like DocuSign, Adobe Sign, Zoho Sign) are developed — with a strong focus on **security, scalability, auditability, and compliance**.

This application demonstrates how Java is used to build **mission-critical backend systems** handling sensitive documents and workflows.



Aim of the Project

The aim of this project is to:

- Build a **DocuSign-like digital signature platform** using Java
- Learn **enterprise backend development with Spring Boot**
- Implement **secure authentication and authorization**
- Handle **PDF processing and digital signatures**
- Design **audit logs and document lifecycle workflows**
- Create a **job-ready enterprise portfolio project**

This project focuses on **real business logic**, not just CRUD operations.

Problems This Application Solves

- Manual document signing is slow and inefficient
- Physical documents lack traceability and security
- No audit trail for compliance verification
- Risk of document tampering after approval
- No centralized document status tracking

The app solves these issues through **digital signatures, immutable PDFs, access control, and audit logging**.



Real-World Use Cases



Real-World Use Cases

1 Corporate & Enterprise

- Vendor agreements
- Employee contracts
- Procurement approvals

2 HR & Recruitment

- Offer letters
- NDAs
- Policy acceptance

3 Legal & Compliance

- Contract approvals
- Regulatory documents
- Client consent forms

4 Finance & Banking

- Loan agreements
- KYC authorization
- Compliance documents

5 Education & Government

- Admission forms

- Certificates
 - Official approvals
-



Industry Value of This Project



Industry Value of This Project

Why Companies Love This Project

This project reflects **real enterprise systems** used in:

- Banking & FinTech
- LegalTech platforms
- Government portals
- HR management systems
- Compliance-heavy industries

Skills Demonstrated

- Spring Boot expertise
- Secure API design
- PDF processing in Java
- JWT & Spring Security
- Audit & compliance workflows
- SaaS architecture understanding



Why This Project Is Resume-Strong

Java-based DocuSign clones demonstrate enterprise backend maturity

This project proves you can:

- Handle sensitive data

- Build secure systems
- Design scalable APIs
- Implement real business workflows

It's **far more impressive** than typical CRUD projects.



Tech Stack (Java-Based)

Tech Stack (Java-Based)

◆ Backend

- Java 17+
- Spring Boot
- Spring Security
- JWT
- Spring Data JPA (Hibernate)
- PostgreSQL / MySQL
- Maven/ Gradle

◆ PDF & File Handling

- Apache PDFBox (PDF reading & writing)
- iText (optional) for advanced signing
- MultipartFile (file uploads)

◆ Frontend

- React
- Tailwind CSS
- react-pdf
- Drag & Drop (dnd-kit)

◆ Other Tools

- Java Mail Sender (email notifications)

- **Docker** (optional)
 - **Cloud Storage** (AWS S3 / Azure Blob or supabase)
-



2-Week Build Plan (Java Stack)



2-Week Build Plan (Java Stack)



Week 1: Backend Foundation & Core Features



Day 1: Project Setup

- Initialize Spring Boot project
- Configure Maven dependencies
- Setup PostgreSQL / MySQL
- Setup frontend (React + Tailwind)

Skills: Spring Boot architecture, project scaffolding



Day 2: Authentication & Authorization

- User entity (id, name, email, password)
- Password hashing using BCrypt
- JWT-based login & registration
- Secure APIs with Spring Security

Skills: JWT, Spring Security, password encryption



Day 3: File Upload API

- Upload PDFs using MultipartFile
- Save file metadata to DB

- Secure upload endpoint

Skills: File handling, REST APIs



Day 4: Document Listing & Preview

- Fetch documents for logged-in user
- View document details
- Frontend PDF preview using react-pdf

Skills: RESTful design, frontend-backend integration



Day 5: Signature Schema & Logic

- Signature entity (docId, signerId, x, y, page, status)
- Save signature coordinates
- Display signature placeholders

Skills: Entity relationships, coordinate systems



Day 6: Drag-and-Drop Signature UI

- Frontend overlay on PDF
- Capture relative coordinates
- Save via REST API

Skills: UX logic, spatial data handling



Day 7: Buffer & Testing

- API testing (Postman)
 - Fix integration bugs
-



Week 2: Enterprise Features & Deployment



Day 8: Generate Signed PDF

- Embed signature text/image using PDFBox
- Generate final signed PDF
- Prevent document modification

Skills: PDF processing, file immutability



Day 9: Email & Public Signature Links

- Generate token-based public signing URLs
- Send email notifications

Skills: Tokenized access, email integration



Day 10: Audit Trail

- Log actions (upload, view, sign)
- Capture timestamp & IP
- Fetch audit history

Skills: Middleware, compliance logging

July
17

Day 11: Signature Status Flow

- Pending / Signed / Rejected
- Rejection reason handling

Skills: Workflow management

July
17

Day 12: Dashboard UI Polish

- Filter documents by status
 - Responsive Tailwind UI
-

July
17

Day 13: Deployment

- Backend: AWS EC2 / Render / Railway
 - Frontend: Vercel / Netlify
 - DB: RDS / Supabase
-

July
17

Day 14: Final Testing & Demo

- README documentation
 - Demo video walkthrough
 - Sample test accounts
-



Java Project Folder Structure

Java Project Folder Structure

```
/signature-app
|— /backend
|   |— controller
|   |— service
|   |— repository
|   |— model
|   |— security
|   |— dto
|   └─ SignatureAppApplication.java
|
|— /frontend
|   |— /src
|   |   |— components
|   |   |— pages
|   |   └─ utils
|
|— application.yml
|— README.md
```

API Endpoints

Auth APIs

POST /api/auth/register

POST /api/auth/login

Document APIs

POST /api/docs/upload

GET /api/docs

GET /api/docs/{id}

Signature APIs

POST /api/signatures

GET /api/signatures/{docId}
POST /api/signatures/finalize

Audit APIs

GET /api/audit/{docId}



Learning Resources – Java DocuSign Clone



Learning Resources –



How to Use These Resources

- Don't watch everything end-to-end
 - Use them **day-wise**, only when that feature comes up
 - Build → get stuck → watch → implement
-



1. Java + Spring Boot Fundamentals



YouTube

Spring Boot Full Course (Beginner to Intermediate)

👉 <https://www.youtube.com/watch?v=vtPkZShrvXQ>

✅ Covers:

- Controllers
- REST APIs
- Dependency Injection
- Project structure

Spring Boot REST API Tutorial (Practical)

👉 <https://www.youtube.com/watch?v=9SGDpanrc8U>

✅ Perfect for:

- `/api/docs`
 - `/api/signatures`
 - `/api/audit`
-



Documentation

- Spring Boot Official Docs
👉 <https://spring.io/projects/spring-boot>
-

2. Authentication & JWT (VERY IMPORTANT)

YouTube

Spring Boot JWT Authentication (BEST ONE)

 <https://www.youtube.com/watch?v=X80nJ5T7YpE>

 Covers:

- Register & Login
- JWT creation
- JWT filter
- Securing routes

Spring Security + JWT Full Tutorial

 <https://www.youtube.com/watch?v=KxqIJblhzfl>

 Enterprise-style security setup

Documentation

- Spring Security Docs
 <https://spring.io/projects/spring-security>
 - JWT (jjwt library)
 <https://github.com/jwtk/jjwt>
-

3. File Upload (PDF Upload API)

YouTube

Spring Boot File Upload (MultipartFile)

 <https://www.youtube.com/watch?v=YijgGq3Fqys>

 Exactly for:

- `/api/docs/upload`
 - PDF uploads
 - Saving files on server
-

Documentation

- Multipart File Upload (Spring)



<https://docs.spring.io/spring-framework/docs/current/reference/html/web.html#mvc-multipart>

4. PDF Reading & Writing (SIGNATURE CORE)

YouTube

Apache PDFBox Full Tutorial

 <https://www.youtube.com/watch?v=4R5S3uP8b1Q>

 Covers:

- Reading PDFs
- Writing text
- Adding images
- Editing pages

Add Image/Text to PDF using PDFBox

 <https://www.youtube.com/watch?v=6RdbzXzqQyw>

 Perfect for embedding signatures

Documentation

- Apache PDFBox Official Docs

 <https://pdfbox.apache.org/>

5. Digital Signature Logic (Coordinates & Embedding)

YouTube

How Digital Signatures Work (Conceptual)

 https://www.youtube.com/watch?v=GSIDS_lvRv4

 Understand:

- Why coordinates matter

- How signatures are rendered

PDFBox Positioning Text & Images

👉 <https://www.youtube.com/watch?v=E9nGJ3Z9N8Q>

✅ Needed for:

- x, y positioning
 - Page number logic
-

6. Audit Logs & Middleware

YouTube

Spring Boot Filters & Interceptors

👉 <https://www.youtube.com/watch?v=QxHk9xYz5vw>

✅ Used for:

- Logging IP
 - Logging timestamp
 - Audit trails
-

Documentation

- Spring Interceptors



<https://docs.spring.io/spring-framework/docs/current/reference/html/web.html#mvc-handlermapping-interceptor>

7. Email & Public Signing Links

YouTube

Send Email Using Spring Boot (JavaMailSender)

👉 https://www.youtube.com/watch?v=QqP_0v5cB8Q

✅ Used for:

- Sending signature links
- Email notifications

Documentation

- Spring Email Docs
👉 <https://www.baeldung.com/spring-email>

8. Testing APIs (Postman)

YouTube

Postman API Testing for Beginners

👉 <https://www.youtube.com/watch?v=VywxIQ2ZXw4>

✅ Use for:

- Auth testing
- File upload testing
- Signature APIs

9. Frontend (React + PDF + Drag & Drop)

YouTube

React PDF Viewer Tutorial

👉 <https://www.youtube.com/watch?v=Uu4W8r9sKxQ>

✅ Used for:

- Viewing PDFs
- Overlaying signature boxes

Drag & Drop in React (dnd-kit)

👉 <https://www.youtube.com/watch?v=FZt2N6R0YkY>

✅ For:

- Signature placement

10. Deployment (Optional but Valuable)

YouTube

Deploy Spring Boot App (Render / Railway)

👉 <https://www.youtube.com/watch?v=1h4XxW6XnGQ>

Deploy React App on Vercel

👉 <https://www.youtube.com/watch?v=2h6n5Dgk0ol>

Recommended Learning Order (VERY IMPORTANT)

Step	Learn
1	Spring Boot REST APIs
2	JWT + Spring Security
3	File Upload
4	PDFBox
5	Signature Coordinates
6	Audit Logs
7	Email
8	UI Polish
9	Deployment

Why These Resources Are Enough

These resources cover 100% of what real enterprise Java apps use

If you complete this project using these links:

- You are **job-ready for Java backend roles**
- You can confidently explain **security, files, PDFs, and workflows**
- Your portfolio will stand out massively